

Skills Summary

Unit Circle — Degree-Radian and Quadrant-Sign Errors

Use this reference while completing your worksheet or when studying.

Skill 1 — Reference angle, then quadrant sign

Two-part rule: an exact value is (*value of the reference angle*) with (*the quadrant's sign*).

- Reference angle: the acute angle to the nearest part of the x -axis. In quadrant II use $\pi - \theta$; in III, $\theta - \pi$; in IV, $2\pi - \theta$ (degrees: $180^\circ - \theta$, $\theta - 180^\circ$, $360^\circ - \theta$).
- Signs (ASTC): quadrant I all +; II only sin +; III only tan +; IV only cos +. Cosine reads the x -coordinate and sine the y -coordinate, so the signs are just the signs of the coordinates.

Example: Find the exact value of $\cos \frac{5\pi}{6}$.

Step 1. The angle is between $\frac{\pi}{2}$ and π , so it sits in quadrant II where cosine is negative — decide that before looking anything up.

Step 2. Reference angle $\pi - \frac{5\pi}{6} = \frac{\pi}{6}$, and $\cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}$; attach the quadrant's minus sign.

$$\cos \frac{5\pi}{6} = -\frac{\sqrt{3}}{2}$$

Try it: Find the exact value of $\sin \frac{5\pi}{4}$.

check: $-\frac{\sqrt{2}}{2}$

Skill 2 — Getting the angle mode right

Conversions:

$$\text{degrees} \rightarrow \text{radians} : \times \frac{\pi}{180} \quad \text{radians} \rightarrow \text{degrees} : \times \frac{180}{\pi}$$

Size check that catches a reversed conversion: a full turn is 360° but only $2\pi \approx 6.28$ radians. Any angle under one turn has a radian measure below 6.28; a converted answer in the hundreds or thousands means the factor was upside down.

Mode check: a bare number with no degree symbol means radians. If a calculator returns a sign the quadrant forbids, suspect the mode before the mathematics.

Example: Convert 300° to radians.

Step 1. Going from degrees to radians, so multiply by $\frac{\pi}{180}$ — the factor that makes the number smaller.

Step 2. $300 \cdot \frac{\pi}{180} = \frac{300\pi}{180}$, and dividing both parts by 60 gives the simplified form. $\frac{5\pi}{3}$, about 5.24 radians — under 2π ✓

Try it: Convert 135° to radians, in exact form.

check: $\frac{3\pi}{4}$

Skill 3 — Solving on an interval

Method: isolate the trig ratio, take the reference angle from the **size** of the ratio, then use the **sign** to choose the quadrants — never the other way round.

- $\sin \theta > 0$: quadrants I, II. $\sin \theta < 0$: III, IV.
- $\cos \theta > 0$: quadrants I, IV. $\cos \theta < 0$: II, III.
- Expect **two** solutions per full turn for a sine or cosine equation (one at each of two quadrants), and answer in the unit the interval uses.

Example: Solve $2 \sin \theta - 1 = 0$ for $0^\circ \leq \theta < 360^\circ$.

Step 1. Isolate first: $\sin \theta = \frac{1}{2}$ is positive, so the solutions are in quadrants I and II — the sign picks the quadrants, the size picks the reference angle.

Step 2. Reference angle 30° ; quadrant I gives 30° and quadrant II gives $180^\circ - 30^\circ$. $\theta = 30^\circ$ and $\theta = 150^\circ$

Try it: Solve $2 \cos \theta + 1 = 0$ for $0^\circ \leq \theta < 360^\circ$.

check: $\theta = 120^\circ, \theta = 240^\circ$

(!) Watch out: A mode error and a quadrant-sign error both return a number of believable size, so recomputing never exposes them. Predict the sign from the quadrant *before* you evaluate, and reject any answer that contradicts the prediction.